



Tactical moves in a fleet

When you're racing in a big fleet, or even a smaller fleet that is pretty competitive, one of the biggest challenges is being able to sail in the direction you want with clear air. The problem is that other boats are constantly infringing on your space; they either have the right of way or their wind shadows slow you down.

The ability to follow a strategic plan within a fleet of boats is an art that requires the ability to think ahead and a solid repertoire of tactical moves. It's tough enough to handle one other boat at a time, but you often must deal with several boats simultaneously. This presents a multi-dimensional challenge that requires good 'fleet management' and is the subject of this issue of *Speed & Smarts*.



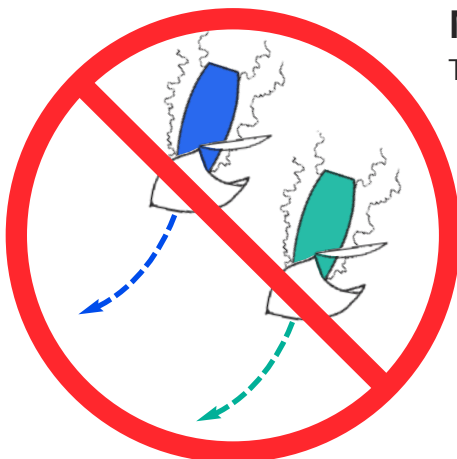
TACTICS...

... are racing maneuvers you make because of the presence or position of one or more other boats. The main purpose of tactics is to help you stay in control of your race and stick to your gameplan. Tactics can also be used when you are trying to catch or stay ahead of one or more specific boats.

ISSUE #143

Fleet Tactics

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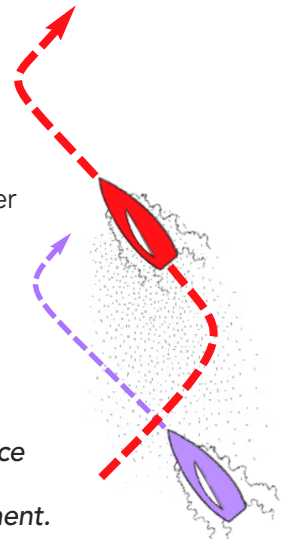


TACTICS = SLOW

Minimize your tactical plays

This entire issue is about helping improve the tactical moves you make relative to the boats around you. But the goal in any race should be to eliminate tactics altogether, or at least to minimize how much you use them. Tactics are slow because whenever you maneuver around one other boat you lose time and distance to the rest of the fleet. So learn the best tactical moves possible, but use them only when necessary.

► The amount of distance you lose in a tack depends on the wind and sea conditions, of course, plus the maneuverability of your boat and talent of your crew. What's clear is that whenever you tack you lose distance to every other boat, so think twice before using this maneuver for tactical purposes like covering an opponent.



Tactical principles

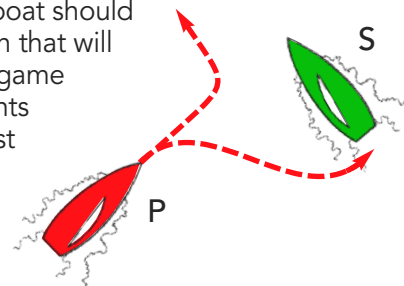
Tactics, like most aspects of sailboat racing, are usually most successful when they are well-thought-through, consistent and done with the big picture in mind. A race might involve hundreds of tactical decisions by each boat, but it won't go well if these are made independently, using inconsistent criteria. Of course, every race situation is different and calls for a unique tactical response. However, you'll have the best results when your decisions are made within the context of a broader philosophy. For example, you might decide to: use minimal tactics, avoid other boats as much as possible, never match race until near the finish, avoid tacking on other boats unless it's necessary, and so on.

Strategy first, then tactics

Two racing boats, P and S, are converging on opposite tacks on a beat. P cannot cross in front of S, so what should P do: Tack into a leebow position on S, or bear off and pass behind S?

The answer depends, of course, on P's strategic plan at that point. If she favors the left side of the course, she should tack; if she likes the right side she should duck.

In almost all cases, a boat should choose the tactical option that will be best for her strategic game plan. The last thing P wants to do is pick the tactic first (e.g. a leebow tack) and then be forced to go where that takes her.



Not all tactics are created equal

Tactics are maneuvers you make relative to other boats, including lee-bow tacks, ducks, covering tacks, luffs and so on. All of these moves are used to help you achieve certain goals on the race course, but they don't all have the same purpose.

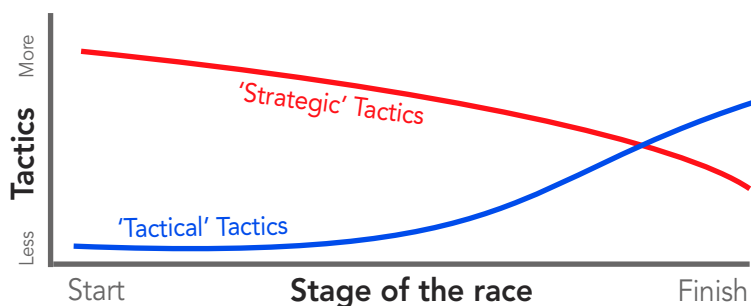
There are two general types of tactics. The first, and by far most common, are 'strategic' tactics. The second can be described as 'tactical' tactics.

'Strategic' Tactics – These are the maneuvers you make relative to other boats for the purpose of following your strategic game plan. Without strategic tactics it would be difficult to follow your strategy because you would get pushed around by the rest of the fleet. Tactics, in this case, are used to maintain control over where you go on the race course. Strategic tactics are valuable tools at any time when strategy is key to success. This includes just after the start of a race, near the beginning of any leg or any time when wind shifts or puffs are producing big gains and losses. Using strategic tactics is seldom a bad idea.

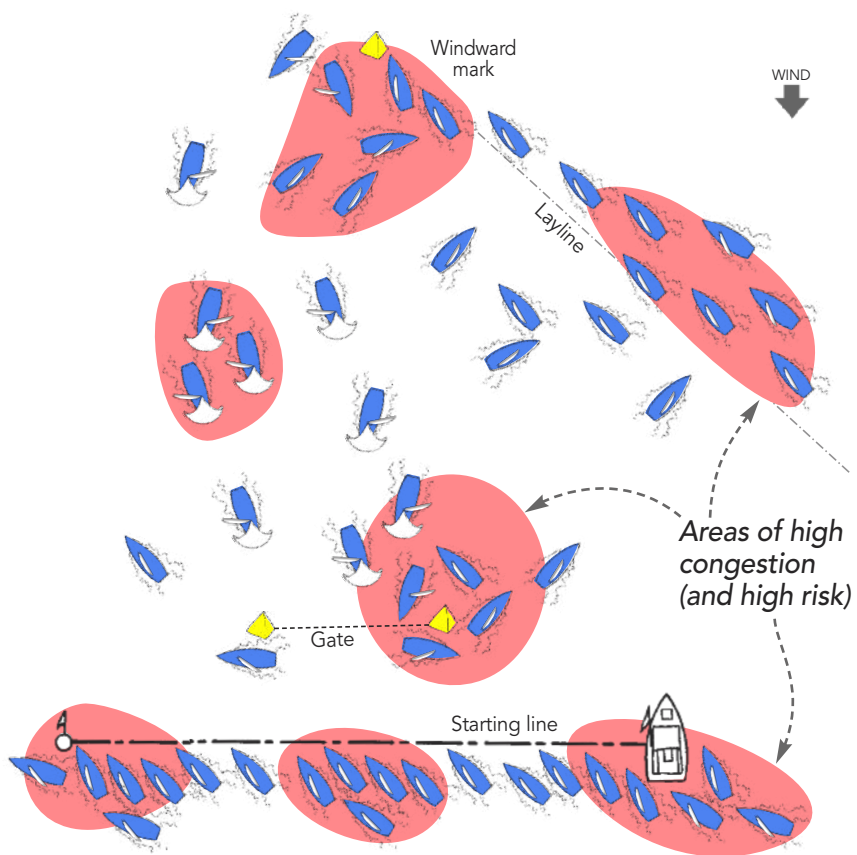
'Tactical' Tactics – These are maneuvers you make relative to another boat(s) for the purpose of passing or staying ahead of them. For example, as you approach the finish you might put a tight cover on a boat close behind you to make sure you beat them in the race. Tactical tactics are usually reserved for the last part of a race or leg (where it can be critical to beat a boat around the mark), but don't get caught using these early in a race or leg when strategy is more critical than tactics. If in doubt about whether or not to use 'tactical' tactics, don't.



A large majority of the tactical maneuvers you make during a race should be to help you follow your strategic game plan. A smaller percentage (maybe 10 - 20%) are used to help you pass or stay ahead of one or more particular boats, usually just before you get to a mark or to the finish line.



◀ The nature of your tactical moves depends, among other things, on where you are between the start and finish of a race. In the early stages, most or all of your tactical moves should be strategic; i.e. their purpose should be to get you to the next windshift or puff. This is not the time to worry about specific other boats. As you get closer to the finish, however, the boats are more spread out and the cost of tactical moves is not so great. This is when you can afford to maneuver tactically to beat nearby boats (see page 9 for more).



Avoid other boats!

A big key to success in almost every race is following your strategic plan – by playing the shifts, finding the best pressure and so on. Being near other boats will almost never help you do this. Your competitors get in the way by casting wind shadows and blocking your path, so an important tactical rule of thumb is simply to avoid the rest of the fleet as much as possible. Look ahead for places on the race course where there will likely be an above-average density of boats (e.g. at the ends of the starting line, near laylines and close to marks), and try to stay away from those crowds. You don't need to sail off by yourself, but keeping a bit of space around you will minimize the need for costly tactical maneuvers and give you a better chance of sticking to your game plan.

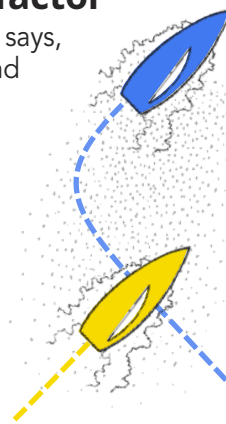


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The karma factor

As an old proverb says, "What goes around comes around." It means a person's actions will have consequences for that person.

In sailing, I believe there is tactical karma. For example, if you tack on another boat, they are more likely to tack on you in the future. So my personal guideline is never to tack on (or otherwise hurt) my competitors unless it is essential at that point in the race. In most cases I go out of my way to find places to tack that won't put other boats in my wind shadow. I'm sure this helps me when other boats think about tacking on my air.



SPEED&Smarts

Speed & Smarts (ISSN 1075-5772) is published by:

Four Winds Inc.

PO Box 435, Easton, CT 06612 USA
800-356-2200 or 203-445-0734

E-mail: SpeedandSmarts@optonline.net

Web site: www.SpeedandSmarts.com

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Speed & Smarts is published bi-monthly, issues are numbered sequentially, and issue dates are approximate.

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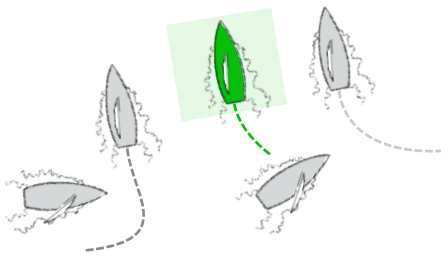
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Issue 143 - Fleet Tactics

Fleet tactics at the start

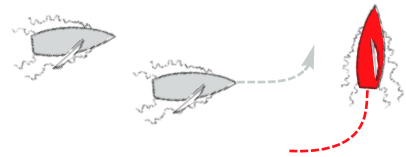
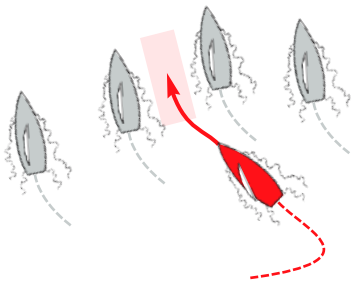
The starting line represents the epitome of fleet tactics. It's the place on a race course where all the boats are packed most tightly together. As a result, every boat has numerous tactical interactions and must make lots of tactical decisions (most of which involve more than one boat). The key to a successful start is using good tactical moves to stay in control of your own destiny. You also have to synch your tactics with the amount of risk you're willing to take. This is one place where you might be happy to trade a perfect strategic position for one that is safer tactically.



Set up in the front row

If you have trouble finding a good spot on the starting line, try getting into the front row of boats on starboard tack a little sooner. It's usually easier to claim a spot earlier in the sequence and defend it (Green) than to look for a spot closer to the start and try to fit into it (Red).

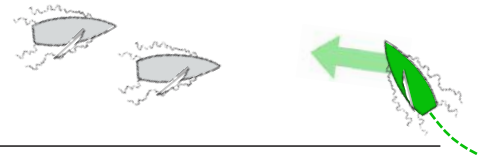
The boats that get into the front row first are able to stake out a certain amount of space around them and then fight to protect it. But the boats that try to squeeze their way into the front row later often have to fit into a smaller space; they end up with boats closer to leeward and windward and not such good odds of getting off the line cleanly. Avoid this by setting up earlier.



Get on starboard sooner

Most boats set up for their start using some version of the port-tack approach. They reach below the line on port tack and then tack to starboard when they find a hole or when it's time to go for the line. The problem is that it's easy for other port tackers to mess up this approach. They can block the port boat from tacking in the first place, or they can luff up underneath the tacking boat (above) and hold her there as long as they want.

To avoid these scenarios and improve your chances of having a good start, get onto starboard tack sooner (i.e. before all the port tackers think about tacking). This will make it easier to defend your position and stay in control of your start (below).



One minute before the start

Pin end

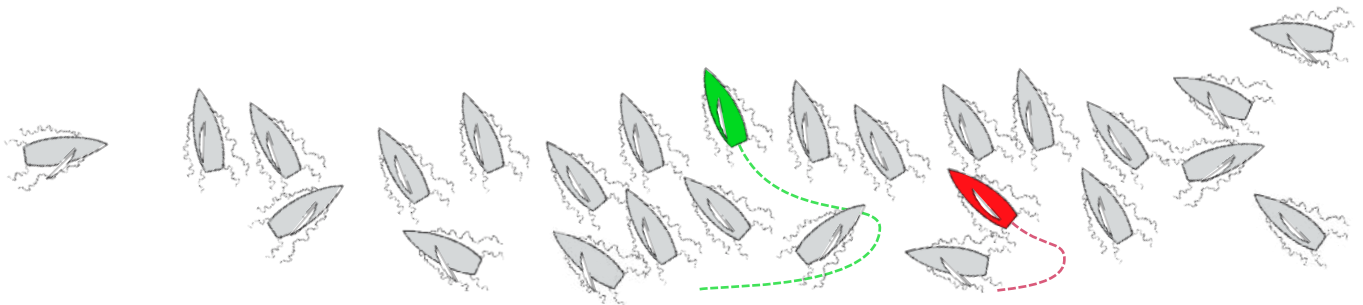
While making your final approach on starboard tack, it's very helpful to be in a front-row position where you have a clear view of both ends of the line and the wind up the course. Being in the front row also gives you options in case the fleet is late to the line (see page 5).

Starting line

WIND



RC boat



GREEN is in a strong position with one minute to go. She's in the front row, poked slightly ahead of the boats around her. This allows her to see both ends of the line as well as her line sights beyond each end. She can also clearly see what the wind is doing up the course. Even if the fleet is late (due to current, a lull, a left shift, etc.) she will be able to sail fast to the line.

RED is not in a strong position at this point. She's in the second row and may have trouble getting to the front of the pack. She has a bad (or non-existent) view of the starting line ends, so she doesn't know far the fleet is from the line. She can't see what the wind is doing up the course, and if the fleet is late to the line she could be in serious trouble.

20 seconds before the start

Looks like there will be a mid-line sag

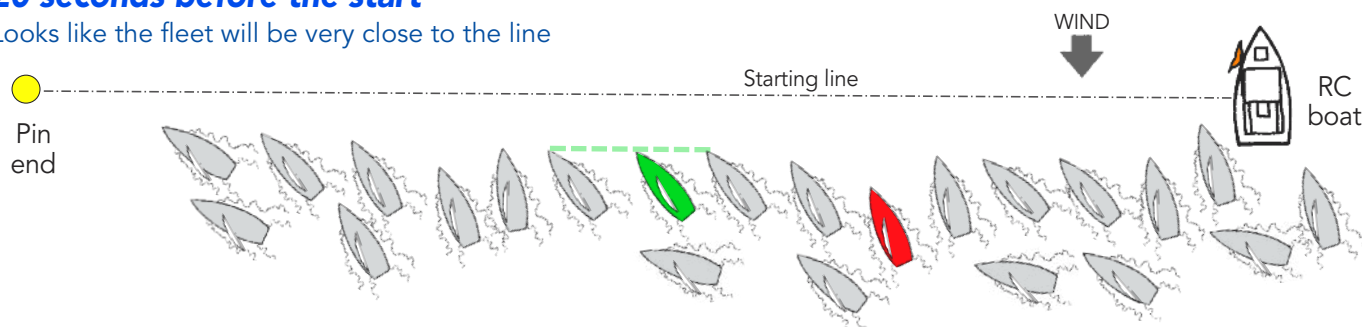


ABOVE: As the fleet approaches the starting line, stay in a poked-out position as long as possible (Green). Being in the front row allows you to see the ends of the line and judge how far the fleet is from the line. This is especially important when you think there may be a sag in the middle of the line. If you don't have a clear view of the line (Red), it will be hard to estimate where you are relative to the line – this is a big reason why boats in the middle are often late to the line. A front-row spot also allows you to sail to the line without interference (e.g. bad air). This is critical when you get close to the start and it looks like there will be a mid-line sag.

BELOW: When starting near the middle of the line, keep your bow in front of the pack so you have a good view of the ends. However, if it looks like the fleet will be on time or early to the line (which could happen if you get a puff, a lift or there is favorable current), don't stay in a poked-ahead position. As boats get closer to the line, slow down and drop back into the line-up. You still want to be in the front row to get clear air at the start (unless you're sure the fleet is over the line), but don't get your bow ahead of the boats on either side – if you do, you will take too great a risk of being identified by the race committee as OCS.

20 seconds before the start

Looks like the fleet will be very close to the line

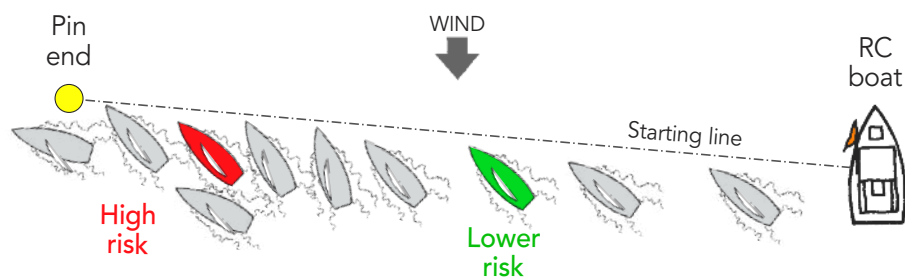


Understanding tactical risk

When sailors talk about risk on the race course, they usually mean risk that is tactical, not strategic. Risk is primarily a tactical thing because the goal of a race is to beat other boats.

When you see a boat way out in one corner of the beat, that may seem like a risky position because of strategic reasons (e.g. it's hard to play shifts from there). But the real risk for that boat is the fact that she is completely on one side of the fleet. If all the other boats were in the corner with her, she wouldn't be taking any risk at all.

Therefore, it's important to know where you are positioned in the fleet. How many boats are on your left side and how many are to your right? This has a lot to do with the riskiness of your position. The key is knowing how much risk you're willing to take at any moment and then playing your tactical cards accordingly.



Improve consistency with conservative tactics

Success in big and/or competitive fleets often depends on being consistent. It's usually better, for example, to finish all your races in the top 7 or 8 than to win a couple of races but get some scores in the teens and twenties. If your scoreline typically looks like a yo-yo, change this by putting yourself in positions that are more conservative tactically.

In order to be consistent, don't always go for perfection. If you continually try to put yourself in the best position, you will probably be taking too much risk. This may work in the short run (a race or two), but it usually ends up producing a yo-yo in the long run (a regatta).

Instead, be willing to make tactical compromises. The starting line is a great place to see the value of this approach. When the pin end is favored and the left side is strong, it's tempting to go for a 'perfect' start at the pin (Red). But this is risky. In the long run, it usually pays to start farther up the line above the pack at the pin (Green). Green won't 'win' the start, but she will have a better chance of continuing to sail toward the favored left side (and therefore a better chance of being in good position at the top mark).

Assess and manage your risk level

In order to manage tactical risk, you need to understand the factors that determine risk. From a fleet perspective there are at least three tactical influences, all related to the location of other boats:

- How many boats are on each side of you? Are there 20 boats on your left and 20 on your right? Or 39 on your left and 1 on your right? This has a big impact on how you will fare if the wind shifts or one side of the course gets better pressure.

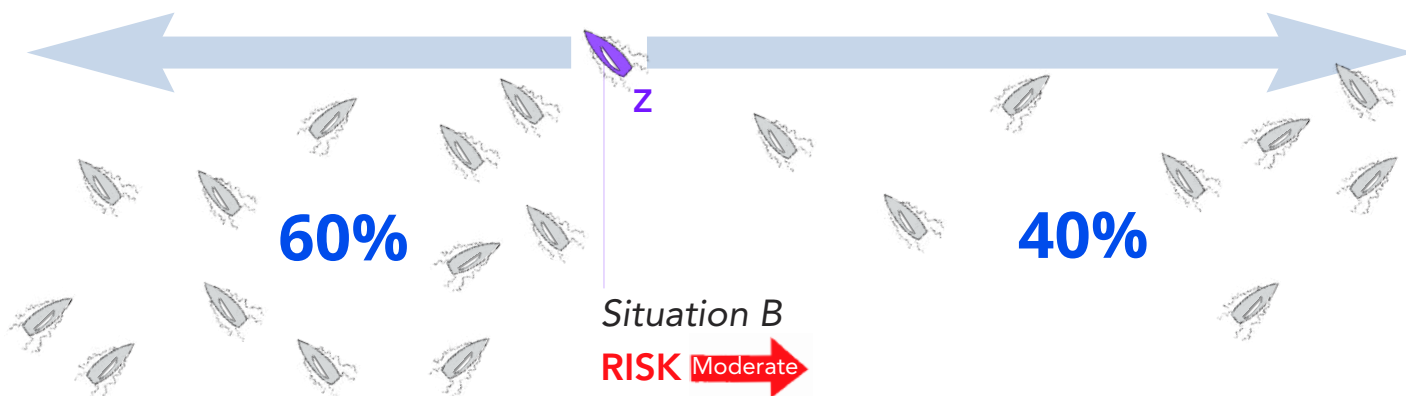
- How far away are other boats? In other words, how much leverage do they have on you? 'Leverage' is lateral separation, or distance measured perpendicular to the wind direction. The farther you are from other boats laterally, the more they will gain or lose when the

wind shifts, so this is a critical tactical concern.

- How close are other boats to you in the race? When you're way ahead of (or behind) the rest of the fleet, your tactical risk is low. But when you are roughly even with a bunch of other boats (i.e. you are all on about the same 'ladder rung'), you are in a risky situation because even a small change can have a big impact.

BELOW: A simple and practical way to measure risk at any point during a beat (or run) is to count the number of boats on each side of you. In Situation A below, Boat Z has 12 boats (60% of the fleet) on her left side and 8 boats (40% of the fleet) on her right. Because Z has allowed 60% (more than half) of the fleet to get to her left side, she has taken a slight amount of tactical risk.



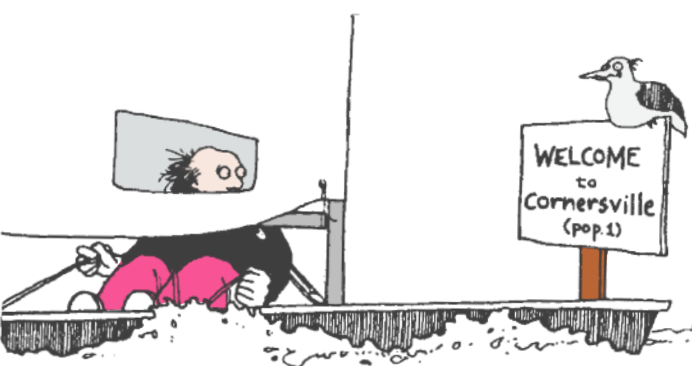
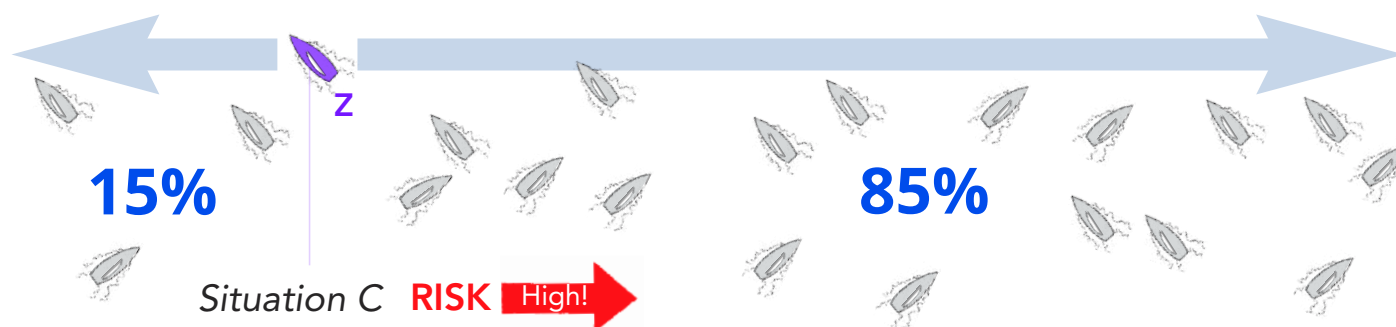


ABOVE: In Situation B, there are 12 boats (60% of the fleet) to the left of Boat Z and 8 boats (40% of the fleet) to her right. However, Z's tactical risk is quite different than it was in Situation A. She did let more than half the fleet get to her left, so that is a slight risk, but most of those boats do not have much leverage on her, and most are well behind Z in the race.

In contrast, the boats to the right of Z are relatively far away (i.e. they have a lot of leverage), and they are quite close to Z in the race (many are essentially on the same ladder rung). These boats pose a greater tactical threat to Z and she is at moderate risk of losing them if conditions change.

BELOW: Situation C shows a very different situation for Boat Z. When her crew takes an assessment of tactical risk on this beat, they count 3 boats (15% of the fleet) to their left and 17 boats (85% of the fleet) to their right. Many of the boats on the right side are close to Z in the race (similar ladder rungs) and they also have a lot of leverage (i.e. they're pretty far away).

This means the risk level for Z is high – even a small shift or pressure change can cost her a lot of boats. There are times when she might be comfortable with this amount of risk (e.g. if she needs to win this race to win the series), but most of the time a good rule of thumb is to keep at least 25% of the fleet on each side.



One of my favorite cartoons from my brother Brad is this perfect illustration of the concept of risk management. Here's a competitor who has sailed all the way out to the right corner of the beat and is now all by himself. There are zero boats to his right, and 100% of his competitors are to his left. This is the ultimate in tactical risk-taking.

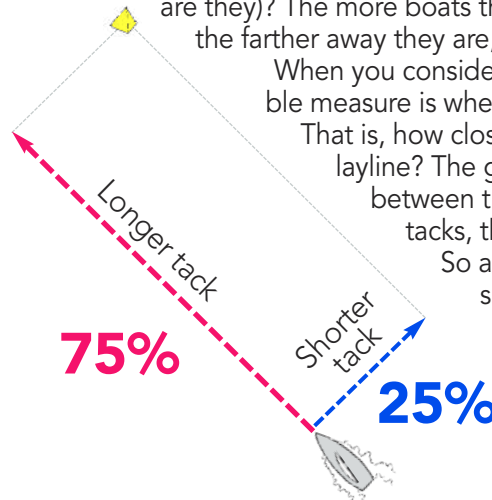
Tactical versus strategic risk

When you're measuring tactical risk, a key factor is lateral position in the fleet. In other words, how many boats are on your left and right sides, and how much leverage do they have (i.e. how far away are they)? The more boats there are on one side, and the farther away they are, the greater the risk.

When you consider strategic risk, a comparable measure is where you are on the course.

That is, how close are you to a corner or layline? The greater the difference between the longer tack and shorter tacks, the more you are at risk.

So a position where you must spend 75% of your time on one tack is riskier than a more central position where you will have to spend only, say, 60% of your time on the longer tack.



Fleet tactics on a windward leg

Racing upwind is often a case of the rich getting richer. The boats that are ahead have clear air and can play the shifts, while the rest of the fleet struggles to find lanes and avoid wind shadows.

Because beats favor the boats ahead, every other boat needs to focus on solid fleet tactics. Some good rules of thumb include:

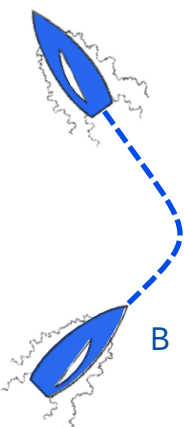
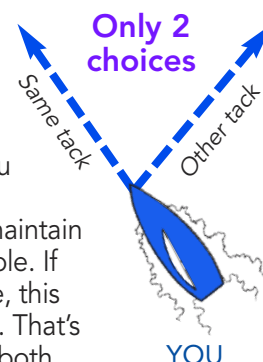
- Keep your options open; don't get pinned going the wrong way.
- Use tactical moves that match your strategic plan (and make sure you have a strategic plan!).
- Don't forget the big picture; try to look 2 or 3 moves ahead.
- Be willing to compromise your strategy for clear air and options.
- Match up your tactics with how much risk you are willing to take.
- Use tactics that are appropriate for where you are on the beat – the beginning, middle or end.

Look for more about each of these guidelines on the pages here and the rest of this issue.

Keep your options open

When it comes to strategy and tactics on a beat (or run), decision-making is binary. That's because at any point in the race you have only two basic choices: 1) you can continue straight on the same tack; or 2) you can tack (or jibe) and sail on the other tack.

Because your options are so limited, it's critical to maintain **both** of these options for as much of the race as possible. If another boat is blocking you from tacking, for example, this means you are giving up 50% of your possible choices. That's not a good thing, so you need tactical moves to keep both options open and stay in control of your own race.



Here's a common tactical maneuver where boats often give up control of their race unintentionally. In this case the Blue boat (B) on port tack could not cross the grey boat (G) on starboard, so B decided to tack into a lee-bow position on G. However, B either tacked too soon, had a bad tack or wasn't advanced enough – as a result she left enough space for G to keep sailing. Now B is pinned on starboard tack. Her weak lee-bow could be a big problem if B suddenly needs to tack (e.g. because another boat tacks directly on her wind).



Don't tack at the front of a 'wedge'

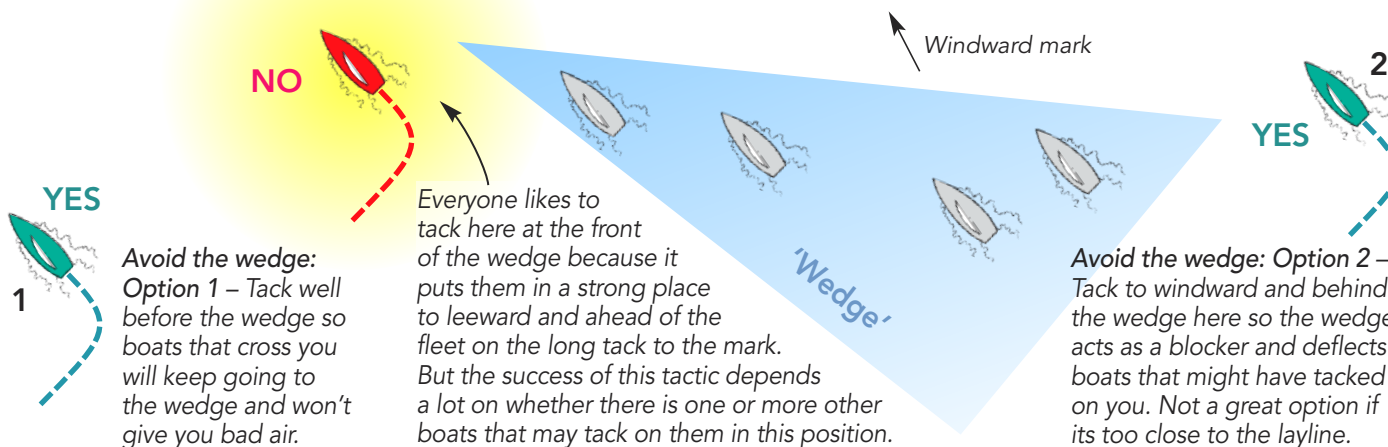
It's something you often find in the middle or second half of a beat: a 'wedge' of three or more boats coming across the course from one side to the other (see below). The boats in a wedge are often stacked up on starboard tack and not too far from the starboard layline, but they could be anywhere, on either tack.

When you converge with a wedge and you want to tack, it's tempting to tack just ahead and to leeward of the lead boat in the wedge (*in the yellow area below*). This can be a strong **strategic** position because you are to leeward and ahead of the pack, usually on the longer

tack to the windward mark, which means you're in a good place to play windshifts.

However, being near the front of a wedge is often a weak **tactical** position because everyone wants to be in that spot. If you're leading the race, this is not a problem, but if there are boats ahead of you, there's a very good chance they will tack on you so they can lead the wedge.

In that case you may want to forget the ideal strategic position and find a better tactical one. By making your tack to leeward of the wedge (#1 below) or to windward of it (#2), you'll stay a little more inconspicuous and may be able to keep your wind clear all the way to the mark.



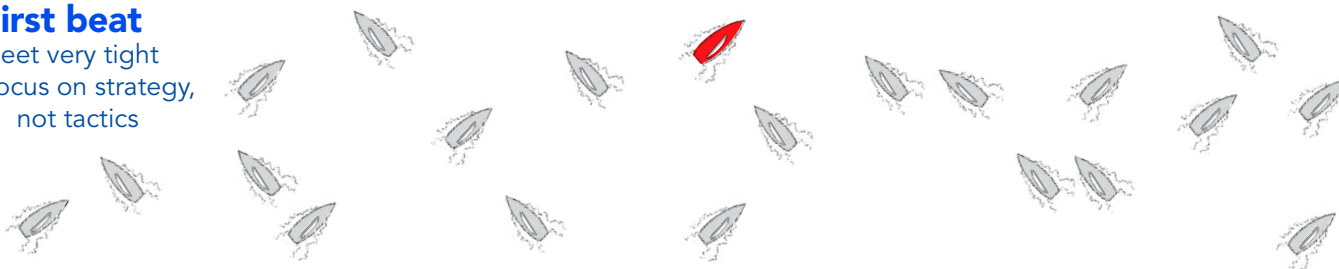
Avoid the wedge: Option 1 – Tack well before the wedge so boats that cross you will keep going to the wedge and won't give you bad air.

Everyone likes to tack here at the front of the wedge because it puts them in a strong place to leeward and ahead of the fleet on the long tack to the mark. But the success of this tactic depends a lot on whether there is one or more other boats that may tack on them in this position.

Avoid the wedge: Option 2 – Tack to windward and behind the wedge here so the wedge acts as a blocker and deflects boats that might have tacked on you. Not a great option if it's too close to the layline.

First beat

Fleet very tight
Focus on strategy,
not tactics



ABOVE: The first beat of a race is usually quite different from the second (or third) beat because the fleet has not yet spread very far apart after the start. With a lot of boats still level in the race, you can't afford to lose distance by dueling with just one or two other boats.

- Focus on strategy as your #1 priority because even small changes in the wind lead to big gains and losses;
- Minimize tactical maneuvers except those needed to follow your strategic gameplan; otherwise don't mess with other boats until close to the windward mark.

Be patient on the first beat and let the fleet sort itself out – then you can start thinking about 'tactical tactics.'

BELOW: By the time boats have sailed the first beat, the run and waited in line to round the leeward mark, the fleet is much more spread out. Sailors should still focus on strategy, especially during the first half of the leg when the boats on each side of the course are still sailing farther apart.

But once boats start converging on the second half of the beat, you can afford to spend more time thinking about the boats around you. Tactical maneuvers still cost the same amount of distance as they did on the first beat, but they aren't as costly in terms of boats lost because there is more space between boats.

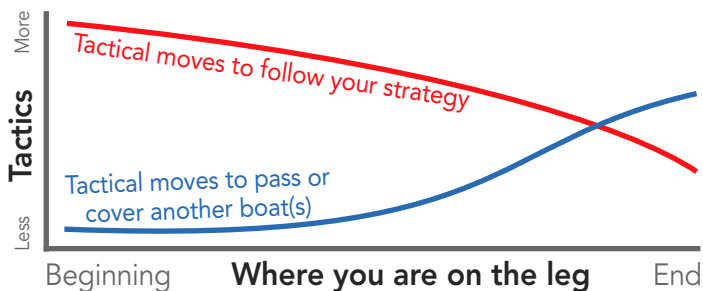
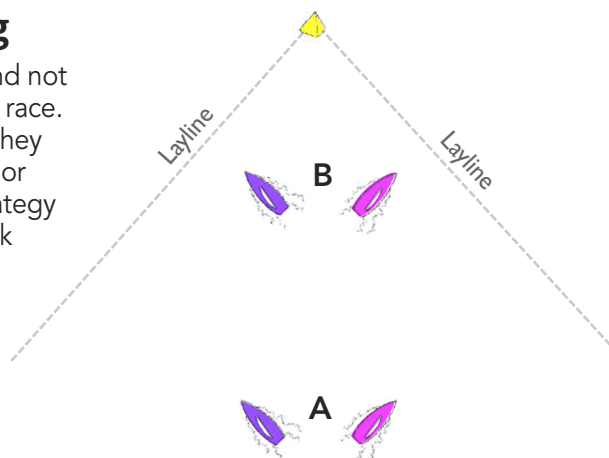
Second beat

Fleet spread apart
Can focus more on tactics
in addition to strategy



Adjust tactics for where you are on each leg

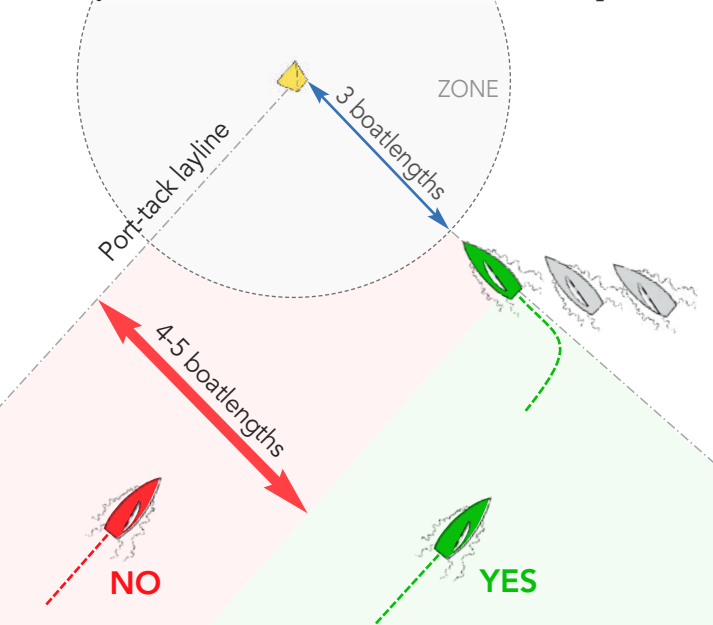
At the beginning of a race you should focus on 'strategic tactics' and not get caught up in 'tactical tactics.' The same is true for each leg of a race. In the early part of a leg (A), boats have a lot of strategic options. They can sail all the way to the left corner, all the way to the right corner or anywhere in between. This is a time when you have to get your strategy right, or you will end up losing a lot. As boats get closer to the mark at the end of the leg (B), however, two things happen: 1) it gets more crowded, and 2) there are fewer strategic options because the course boundaries (i.e. the laylines) converge. This is a time when you need to transition from strategy to tactics so you can get around the mark cleanly and ahead of the nearby boats.



◀ The nature of your tactical moves is affected by where you are on a leg. When you have just begun a leg, most or all of your tactical moves should be strategic; i.e. their purpose should be to get you to the next windshift or puff. This is not the time to worry about specific other boats. As you get closer to the next mark, however, the fleet comes together and there are fewer strategic options. This is when you should focus more on tactical moves to beat nearby boats around the mark.

Approaching the windward mark

The windward mark is the first point where all the boats come back together after the start. It's usually crowded and contentious and can have a significant impact on the outcome of the race. The tricky part is how you approach the mark – the key tactics here are to play by the rules, keep your wind clear and avoid overstanding the mark by too much. This is easy to do when you're near the front (or back) of the fleet, but it requires some smart choices when you reach the mark in the middle of the pack.

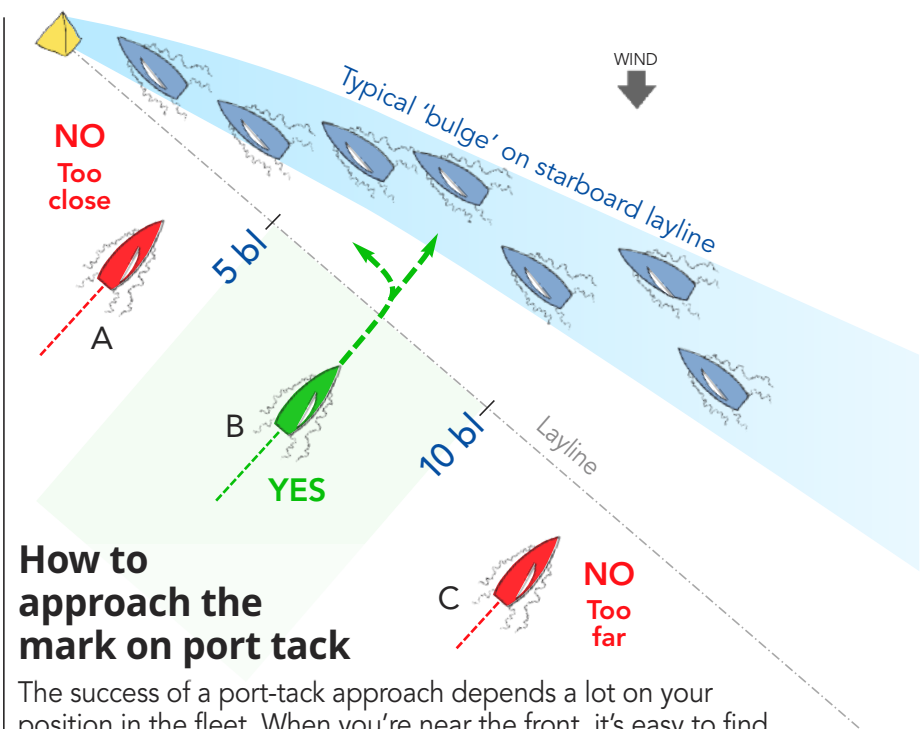


Beware the port-tack layline

I'm not opposed to approaching the windward mark on port tack. In fact, I do it a lot. There are usually fewer boats on the left side at the top of the beat, and this approach is often better than sailing in bad air, overstanding the mark or dealing with other problems found on the starboard-tack layline. But if you come into the mark on port tack you have to do it right; otherwise you risk disaster.

The most important tactical consideration is to avoid being on the actual port-tack layline to the mark. There are times when this approach can work brilliantly, but from a risk-management point of view, the closer you are to the port layline, the bigger your risk.

The primary reason is that the rules are stacked against boats coming in on port. They must keep clear of starboard tackers, of course, and they will have other problems if they tack inside the zone at the mark. Therefore, a basic rule of thumb is to be at least 4 or 5 lengths below the port-tack layline on port tack. This way when you tack near the mark you won't be in the zone and you don't have to worry about rule 18.3 (Tacking in the Zone).



How to approach the mark on port tack

The success of a port-tack approach depends a lot on your position in the fleet. When you're near the front, it's easy to find a hole in the starboard-tack lineup, so you can come in to the mark at almost any distance below the port-tack layline. But when you're in the middle of the pack, the point at which you reach the starboard layline is more critical.

Approach A – Too close to the mark.

When you arrive at the starboard-tack layline, don't be too close to the mark. If you tack from port to starboard and pass head to wind inside the zone, you have a high risk of breaking rule 18.3 (Tacking in the Zone). In addition, the starboard-tack parade is pretty dense and close to the layline when boats get that close to the mark, so it may be impossible to find a gap where you can tack or a lane below the starboard tackers where you can tack and still fetch the mark.

Approach C – Too far from the mark.

When you reach the starboard layline on port tack, there are two places where you can tack: 1) on the windward side of all the starboard tackers, with clear air; or 2) to leeward of the starboard tackers, in bad air. If you're too far from the mark, neither of these is a good option. The starboard tack lineup is usually well above the layline there, so going to windward of them means overstanding by a lot. There is usually room between the starboard tackers and the layline, but tacking there would mean you'd have to sail in bad air for too long.

Approach B – Just about right.

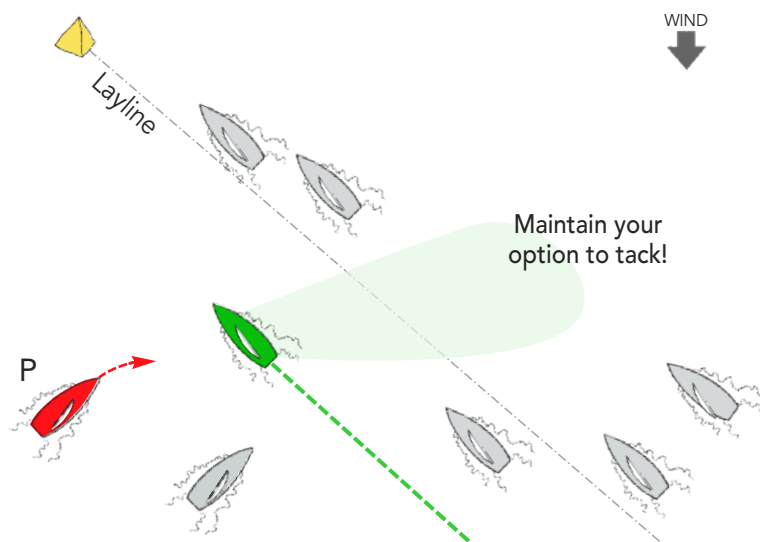
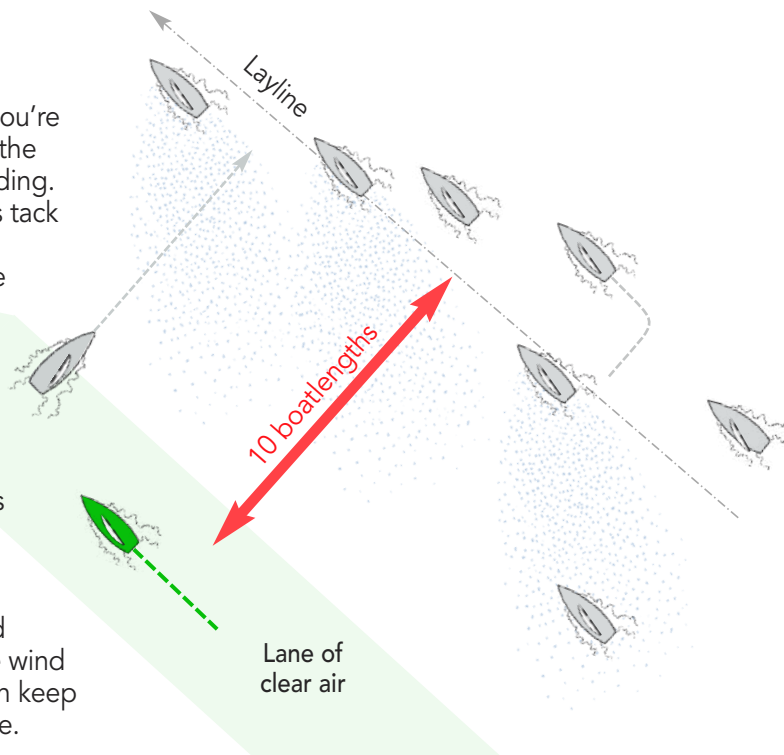
For many fleets and conditions, there's a better compromise between roughly 5 and 10 boatlengths from the mark. This is far enough away so you won't have a problem with rule 18.3, and there is usually room to tack below the starboard tackers and still be above the layline. It's also close enough that you could survive from there to the mark in bad air if necessary.

Warning! Beware of tacking below the line of starboard tackers when you are racing in light air or adverse current.

Try a stealthy 'under-layline'

When you're over on the right side of the beat and you're not winning the race, it can be a challenge to get to the windward mark without sailing in bad air or overstanding. One thing many boats do that clearly doesn't work is tack on or near the layline. The layline is a magnet for all boats in that area. If you tack on the layline and there are boats ahead of you, it's a near certainty that they will tack on your wind. And then you will have to sail in bad air or make two extra tacks and end up overstanding.

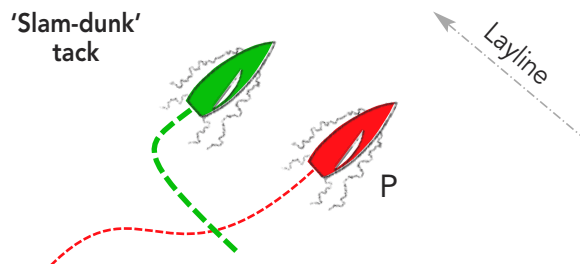
A better idea is to tack in a lane roughly 10 boatlengths short of the starboard-tack layline. The idea is to 'hide' in a place where boats that cross ahead of you will not be likely to tack. Ten lengths from the layline is a good distance for two reasons: 1) most boats that are on port tack and this close to the layline will keep going (and leave you alone); and 2) you will be just far enough to leeward to avoid the wind shadows of boats on the actual layline. Often you can keep clear air all the way to the mark in this low-profile lane.



When you might not be fetching the mark

Sometimes when you're approaching the windward mark it looks like you may fetch the mark on starboard tack, but you're not sure. As long as there's a chance you will make the mark, you'd rather not tack two more times and lose all that distance. But if you end up not fetching the mark, you don't want to wait until it's too late to make the two tacks you need to get around it.

Here is the tactical play: As long as you have the ability to tack (without fouling any other boats), it's OK to keep sailing on starboard tack and give yourself a little more time to see if you will fetch the mark. However, once you become aware of a boat that is a threat to fill the space on your windward side (and block you from tacking), you have to make a quick decision. Do you really think you will make the mark? If so, continue on starboard tack. If not, take proactive measures before you lose the option to tack. This problem usually comes from a port tacker (P) that will bear off and pass behind you. Don't just sit there and let them pin you! At right are two possible moves you can make.

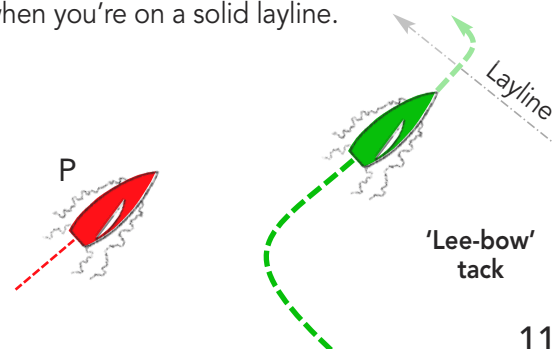


Protect against a port tacker

When you're on starboard tack just below the layline and converging with a port tacker (left), don't let them duck you and then tack on your windward hip. If they do, you may be pinned on starboard and unable to tack to make the mark. There are basically two ways you can prevent this from happening:

Above: A slam-dunk tack. Just after P's bow passes your transom, go into a tack. The goal is to end up close enough to windward of P so you pin *them* from tacking. Then you can tack first once you are fetching the mark. This gives you the best chance of rounding ahead of P.

Below: A lee-bow tack. Before you cross the port tacker, tack to leeward and ahead of them. Do this far enough from P so they cannot pin you from tacking on port. Then tack again when you're on a solid layline.



Downwind fleet tactics

When you're going downwind, you have to shift your tactical focus from the boats ahead of you to the boats behind. Those are the boats that will hurt you with their wind shadows, threaten to get an inside overlap and show you the wind you will soon be getting. But this is not a simple adjustment.

On a beat, it's fairly easy to watch the boats ahead of you because that's where the helmsperson and most of the crew are normally looking. But on a run you have to turn your head and look back. The best solution is to position one crewmember so that he or she is sitting or standing and facing aft. This person's job is to focus on what is happening behind and to windward, and communicate that to the rest of the team so they can keep looking at the waves, spinnaker trim, etc.

Stay away from packs of boats

A key principle of racing in fleets is that tactics are slow, so it's best to stay away from other boats as much as possible. This doesn't mean sailing off by yourself and taking unnecessary risks, but it does mean not getting caught up in pack mentality. On runs it's clear that a boat sailing by herself will almost always be faster than boats sailing together (below). The pack boats are affected by bad air, disturbed wake and rule 10 (Port-starboard). As a result, packs typically perform as a function of their lowest common denominator – moving along roughly at the pace of the slowest boat.

If you find yourself in a pack: 1) Sail extra high or low temporarily to escape the pack; 2) if you can't escape entirely, at least get to the edge of the pack; and 3) get to the edge where you will have clear air on the long jibe to the mark, or the edge where you will be on starboard tack when you converge with the pack.

The boats at the back of the pack are stuck in (and slowed by) the wake of the boats in front of them.

The boats on this side of the pack cannot jibe to port whenever they want because they will have to keep clear of starboard tackers.

Bad air

The Red boat smartly jibes to avoid getting caught in the pack. She's also on the side where, if she must later converge with the pack, she'll be on starboard tack (and inside at the mark).

The boats at the front of the pack are getting bad air from all boats behind them.

Be willing to make strategic-tactical compromises

In the ideal race you always follow your strategic gameplan. If the right corner is strongly favored, for example, you go all the way there because that's the fastest way to get to the next mark. But sometimes executing the perfect strategic plan is not the best idea. The starting line is often a great example of this (see page 6). If the pin end of the line is very favored, strategically you want to start right there. However, you know there will be a big crowd at the pin, so starting there means taking a big tactical risk. Therefore, the best plan may be a compromise between what is best strategically (starting right at the pin) and what is best tactically (starting away from the pack at the pin).

This kind of dilemma comes up all the time on the race course. Consider the typical situation shown here on a run. These boats with asymmetrical spinnakers rounded the windward mark, stayed on starboard jibe and are now heading toward the right corner of the run (looking downwind). You are in the Red boat and your strategy (i.e. what you'd do in the absence of any other boats) says go all the way to the corner before jibing. But here's the risk: If you go to the corner and then jibe, some of the boats behind you might jibe right on your wind, and then you'd have to sail all the way to the leeward mark in bad air. A strategic-tactical compromise would be to jibe early. If you go before all the other boats are thinking about jibing, there's a better chance you will have clear air for the rest of the run, and that often gets you to the leeward mark sooner than following the perfect strategy.

Leeward mark

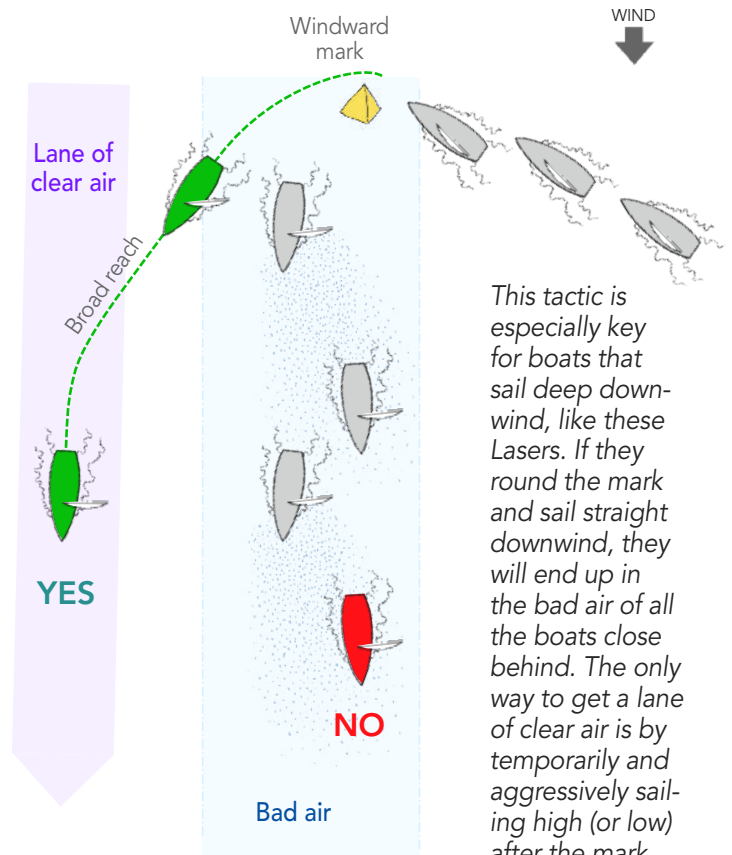
Even when you think it's better to continue sailing toward the corner, it may be smart to bail out early – before all the boats around you decide it's time to jibe. This gives you a better chance of having clear air (after you jibe) on the longer jibe to the leeward mark.

Go for a good lane early

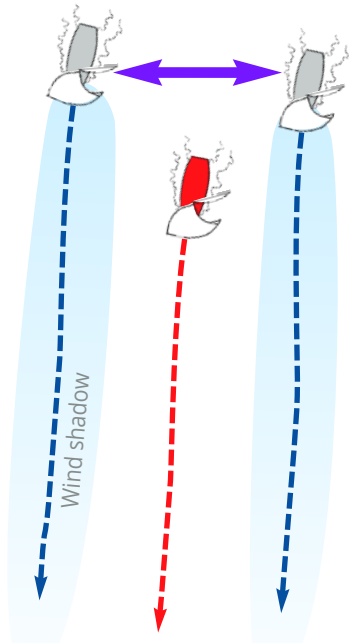
One of the biggest tactical challenges on runs is being able to sail in clear air. This is especially true for boats that sail nearly dead downwind (e.g. singlehanders and boats with symmetrical chutes in moderate or heavy wind) because they don't spread apart like boats do on beats. A boat that rounds the windward mark and sails her optimal downwind angle (almost straight toward the leeward mark) will soon be in the bad air of almost every boat that rounds the mark behind her.

If you don't have many boats behind you, this is not a big problem. But if you approach the windward mark just ahead of a pack, you need a proactive tactical plan. This is another one of those times when the best option may be a compromise between your ideal strategy (e.g. sailing straight downwind) and your tactical need to sail in clear air.

The usual solution is to sail a relatively high angle until you are on a side of the fleet. You can either stay on a broad reach on starboard jibe (right), or do a jibe set and broad reach on port. Go just far enough so you can bear off and not have another boat behind and to windward of you. Then maintain this lane of clear air. As the fleet spreads out down the run, it may be possible to sail back toward the middle with clear air to give yourself more tactical and strategic options.



Flat water



To illustrate how surfing affects wind shadows, these diagrams show a boat with a narrow lane of clear air. But this is not really a strong position because she must sail the same angle as the boats behind, and it's easy to drift into their wind shadows.

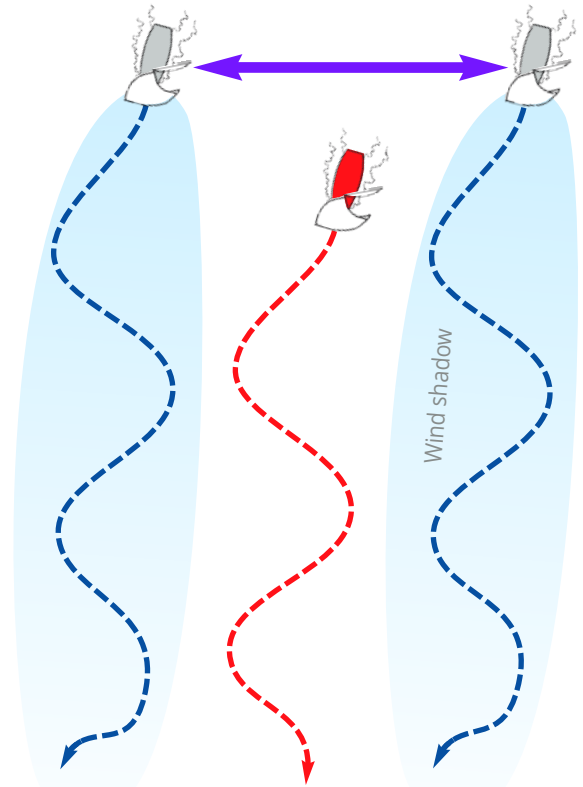
Find a wider lane in surfing conditions

When sailing downwind, the bad air coming from boats behind you is an important tactical factor. It's critical to know the precise location of each boat's wind shadow (it extends to leeward in the direction shown by their telltales) and to avoid this zone of disturbed air.

This is hard enough to do in flat water when boats sail straight courses downwind, but it's even more of a challenge when there is enough wind and waves for boats to surf. When boats are surfing, they sail zig-zag courses (right), so their wind shadows move back and forth constantly.

In these conditions, it's essential to give a wider berth to boats behind so you aren't affected by their wind shadows at any point in your (or their) zig-zagging. When you're trying to catch and stay on waves, the last thing you want is a little less wind pressure – that could be the difference between catching or not catching the next few waves.

Surfable waves



The boats around you also steer up and down a lot when surfing in waves, so consider their paths when trying to keep your wind clear.

Exiting the leeward mark

The more time I spend coaching and watching sailboat races, the more I'm convinced that the leeward mark is the most critical part of any race. It's certainly the place where boats give up the most positions and distance. And it's not hard to do. Even small errors (like getting your bow trapped just to

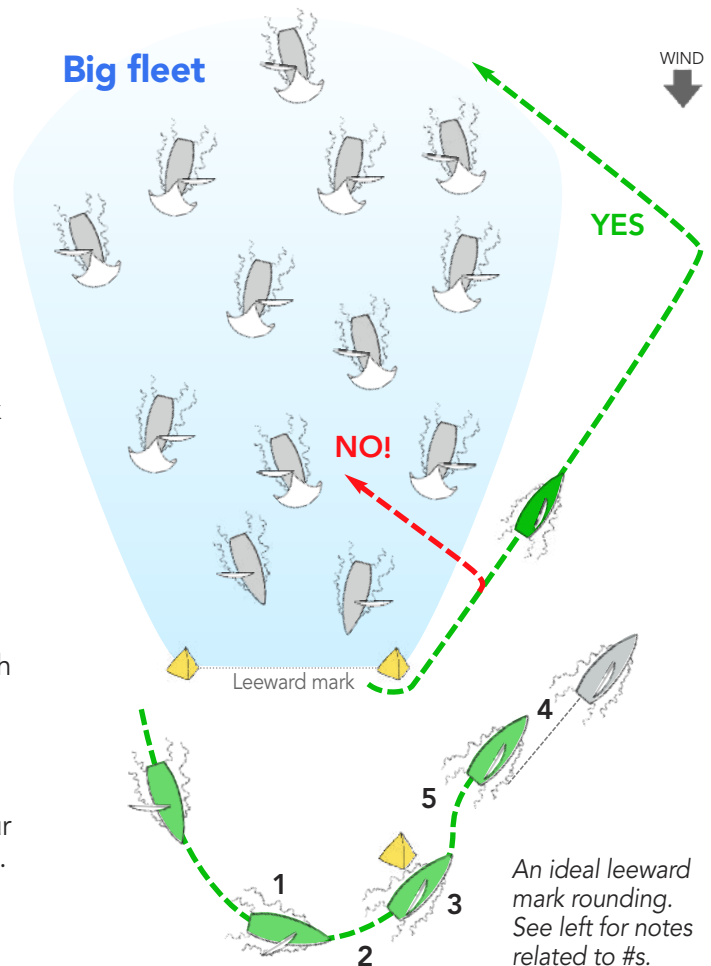
leeward of the boat ahead) can lead to big losses. The way that boats exit the leeward mark has a big effect on their ability to go fast and follow their strategic plan during the first half of the next beat. For these reasons it's well worth figuring out the tactical moves needed to leave the mark with speed and options.

Work hard to hold your exit lane

After you round the leeward mark, it is *always* better if you can 'hold your lane' and keep sailing fast straight ahead as long as possible. This gives you many more strategic and tactical options, especially in a bigger fleet where it can be so tough to find clear air and go the way you want. In fact, the bigger the fleet, the longer you need to be able to stay on the tack you were on when rounding the leeward mark. That's because it is usually not a good idea to tack right after the mark and cross through the fleet coming downwind.

In order to stay in relatively clean air on the same tack after the leeward mark, you need to have a very good rounding. There are five things you should do:

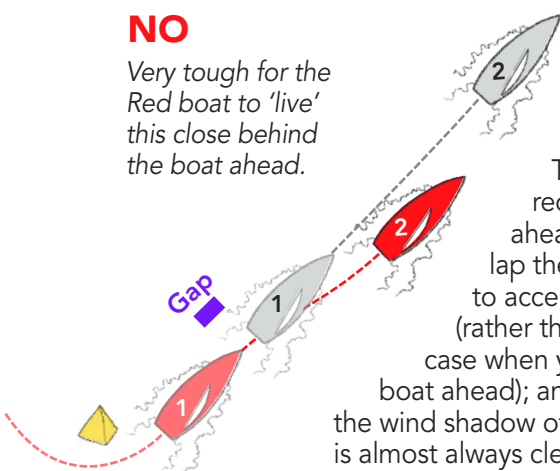
1. *Before* your bow gets to the mark, be 100% ready to race fast upwind. You should not still be putting the spinnaker away or adjusting your sail controls.
2. When your bow gets to the mark, you should be going faster than your normal upwind speed. To do this you have to sail on a beam reach for at least a boatlength just before you get to the mark.
3. Round right next to the mark, close enough so you could reach out and touch it. Don't go wide or outside another boat; slow down if necessary to be at the mark.
4. Have a gap of at least 1/2 boatlength between your bow and the stern of the boat in front of you (see below).
5. Use your extra speed to briefly luff up a little after the mark and climb into a lane that is to windward of the track of the boat ahead of you.



Mind the 'gap' at the mark!

NO

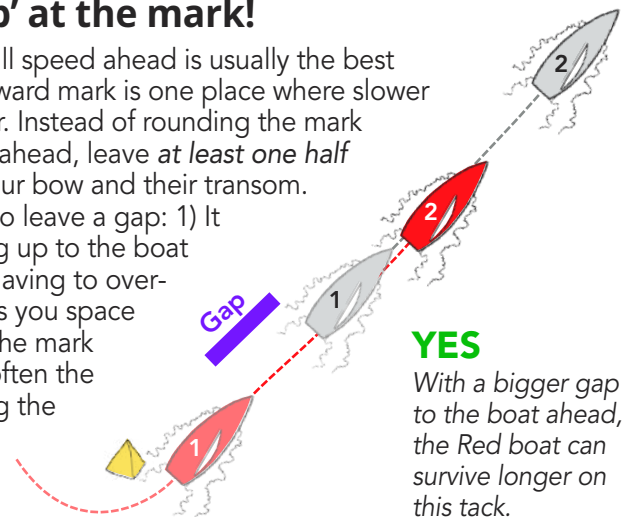
Very tough for the Red boat to 'live' this close behind the boat ahead.



In sailboat racing, full speed ahead is usually the best strategy. But the leeward mark is one place where slower may actually be faster. Instead of rounding the mark right behind the boat ahead, leave *at least one half boatlength* between your bow and their transom.

There are three reasons to leave a gap: 1) It reduces the risk of catching up to the boat ahead and fouling them or having to overlap them to leeward; 2) it gives you space to accelerate as you approach the mark (rather than slow down which is often the case when you worry about catching the boat ahead); and 3) it helps you avoid the wind shadow of the boat ahead. There is almost always clearer air a boatlength behind a boat than behind her transom.

Gap



YES

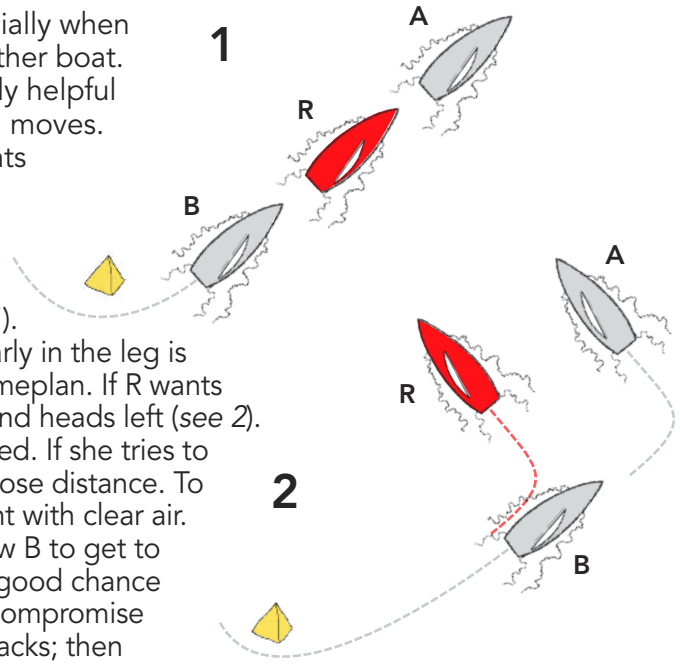
With a bigger gap to the boat ahead, the Red boat can survive longer on this tack.

Options after the leeward mark

Sometimes tactics are a lot like a game of chess, especially when your moves affect (or are affected by) more than one other boat. You don't want to over-complicate things, but it's usually helpful to think ahead about the consequences of your tactical moves. Here's a simple and common situation where three boats round a leeward mark close together. Consider the strategic and tactical moves for the Red boat.

The Red boat rounds second: The Red boat (R) comes around the mark just behind A and just ahead of B (see 1). Unless this is a short beat to the finish, the priority this early in the leg is strategy, so R should use tactical moves to follow her gameplan. If R wants to go left, her next move is easy – she simply tacks first and heads left (see 2).

But if R wants to go right, things are more complicated. If she tries to hang on behind A, she will likely fall into A's bad air and lose distance. To avoid this she can do two tacks and get a lane going right with clear air. But there are a couple problems with this: 1) she will allow B to get to her right (i.e. closer to the favored side); and 2) there's a good chance A will cover R, which may make it harder to get right. A compromise tactic is for R to be patient and stay on port tack until B tacks; then tack. Even if A tacks on R, at least R is doing the same to B.

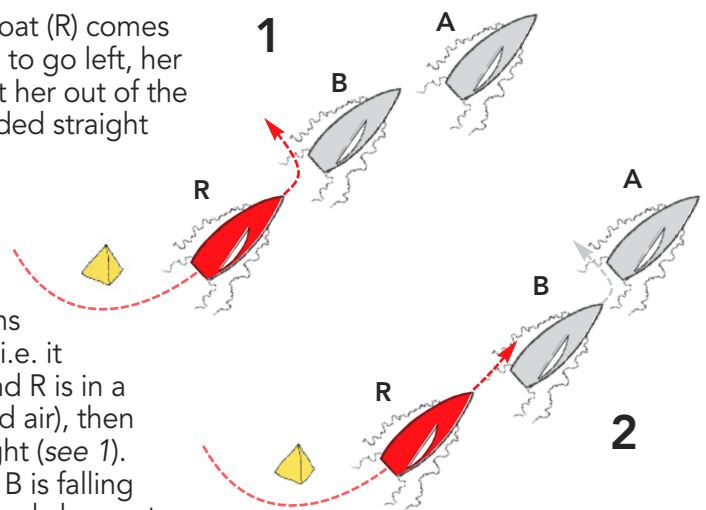


The Red boat rounds third: In this scenario the Red boat (R) comes around the mark behind both A and B. If R's strategy is to go left, her move is clearly to tack as soon as possible. This will get her out of the bad air of the boats ahead and, more importantly, headed straight toward the favored side of the beat.

If R wants to go right, that's a little trickier because she is in the bad air of two boats that are blocking her path to the right. One option is to tack twice and get a clear-air lane to windward of A and B going right.

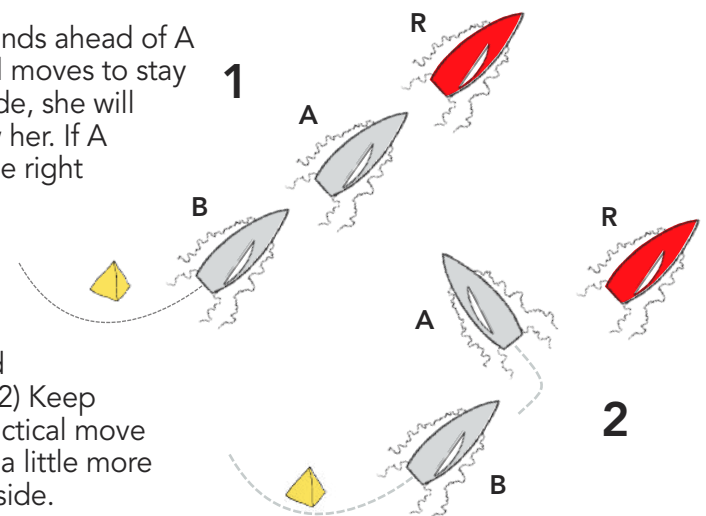
R's tactical choice may hinge on the relative positions of the boats. If B is in a relatively strong position on A (i.e. it looks like B will be able to keep sailing on port tack) and R is in a relatively weak position on B (i.e. R is falling into B's bad air), then two tacks may be R's only way to get clear air going right (see 1).

However, if R had a good rounding and it looks like B is falling into A's bad air (see 2), then R could wait for B to tack and clear out.



The Red boat rounds first: When the Red boat (R) rounds ahead of A and B, she must balance her strategic plan with tactical moves to stay in front of the nearby boats. When R favors the right side, she will obviously keep going as long as the other boats follow her. If A tacks (see 2), R has to make a choice: Keep going to the right side (letting A split but covering B) or tack to go with A (and letting B sail toward the favored side). If R is confident about the right side she will probably keep going; if not, she might tack in a loose cover.

If R favors the left side she has two clear choices: 1) Tack first before either A or B. This is a gutsy call and requires a lot of confidence in her 'go left' strategy; or 2) Keep going right until A tacks, then tack immediately. This tactical move prevents A from getting to the favored side first and is a little more conservative when R is not so confident about the left side.



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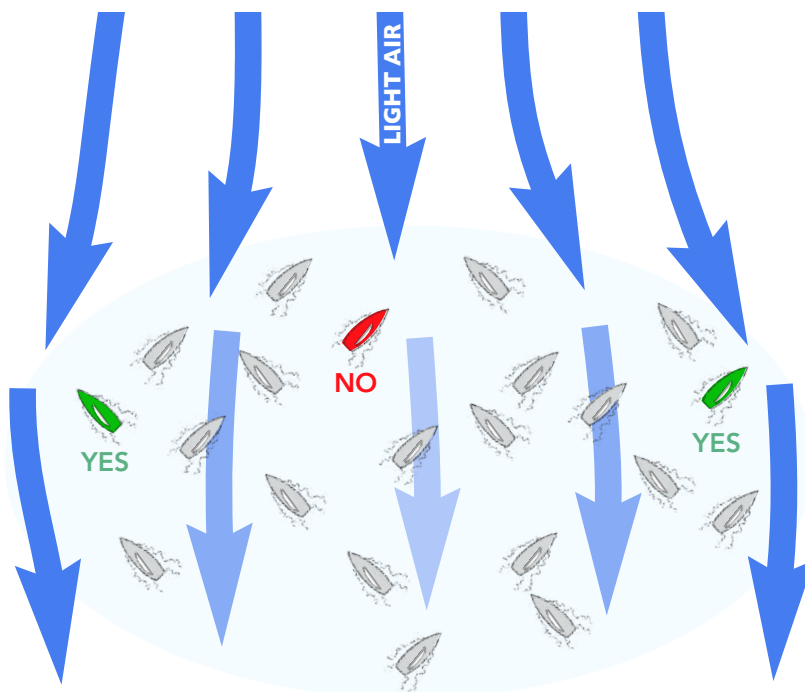
Tactical strategy

Strategy and tactics almost always have the same essential relationship: Strategy is your plan for how you would get around the race course as quickly as possible (using the wind, current, etc.) if there weren't any other boats racing. Tactics are the boat-on-boat maneuvers you make to implement your strategic plan in the midst of a fleet.

But once in a while this relationship flip-flops when the presence of the boats themselves actually becomes a strategic factor and must be considered in your game plan. A great example of this is when you are racing upwind in a big fleet and light air (*right*).

We've all seen how the wind velocity and direction (two huge strategic factors) can be affected when you're racing near a shoreline with buildings and trees. The same thing can happen when there is a lot of boats, especially in light air.

When the wind is strong it will push through the spaces between boats and fill into the middle of a fleet. But in light or even medium air, the wind doesn't have enough energy to fight through all the sails. In these cases it's easier for some of the air to flow over or around the mass of the fleet. That's why you often find more wind pressure at the sides than in the middle, rear or even at the front of the pack (since the wind may start to lift up before it reaches the lead boats).



We know from everyday experience that wind flow is affected by objects in its path. If you're standing behind a tree, for example, you won't feel as much wind as a person standing in an open field. This is obvious, but even if you aren't right behind a tree, you feel less wind if there are a bunch of trees around you. That's because the wind sees the group of trees as one large obstacle, and some of the wind flows over or around that obstacle.

A similar thing happens in a fleet of sailboats. There are wind shadows just to leeward of each boat, but there is also a macro effect from the entire fleet. The array of sails spread out across the water acts like a snow fence in a field – even though there are many gaps, the wind in the middle is noticeably weaker than on the sides. The bigger and more dense the fleet, and the lighter the wind, the more significant this effect will be as a strategic factor.